

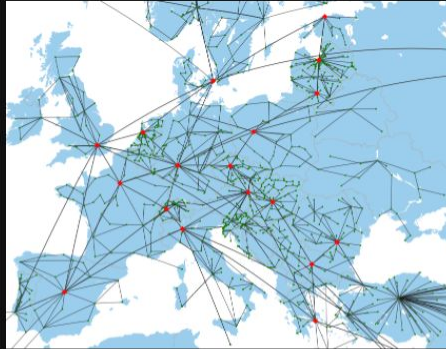
Network Programming

Created by <Academia de Código_>

Computer Networks

Wide Area Network

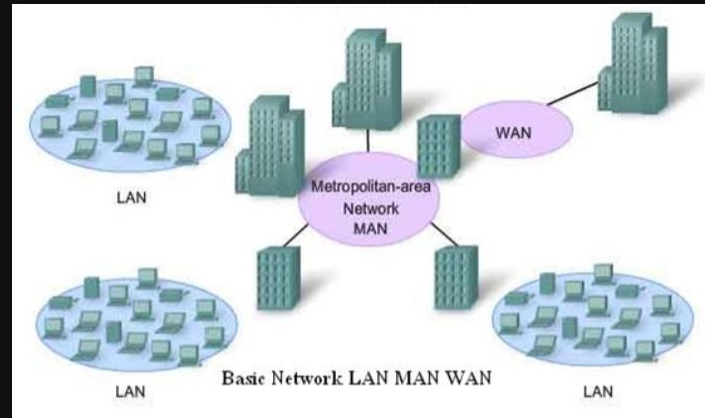
- Communication over large distances, spanning entire continents and joining computers across an ocean;
- Operates at slower speeds and exhibits significant delay;
- Topology consists of a mesh of individual point-to-point links.



Internet Backbone in Europe

Local Area Network

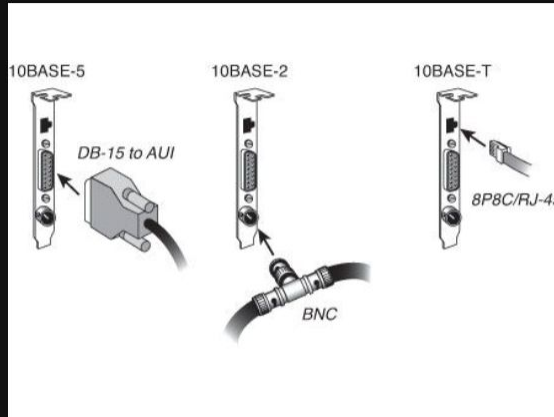
- Short distances only;
- Operates at high speeds and does not exhibit significant delay;
- Each computer contains a network interface that connects the machine to the network medium.



Ethernet

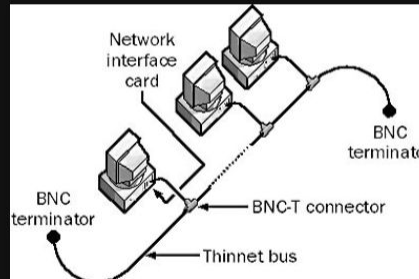
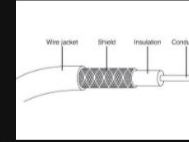


Invented at Xerox in the 1970s, and has become the most popular means for connecting computers in a LAN. Evolved from coaxial cable to twisted pair and fiber optics. Transmission speeds range from the original 10 Mbit/s to current 100Mbit/s, 1Gbit/s and 10Gbit/s.



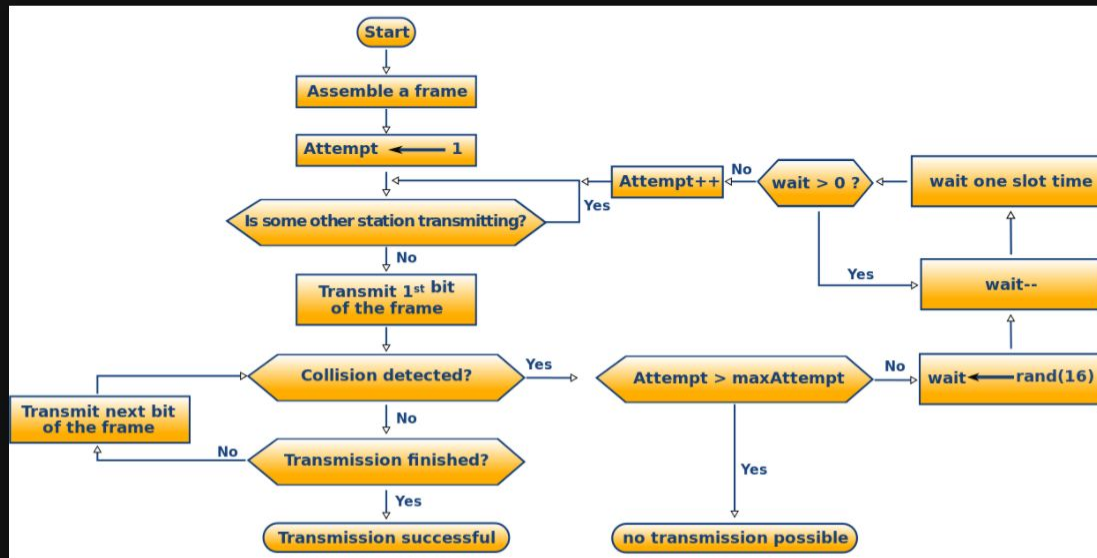
Bus Topology

- All computers share a single communication channel;
- Broadcast-based, all computers receive all transmissions;
- Each computer discards transmissions destined to other computers;
- Collisions can occur and need to be handled;
- 10 Base 5 for 10 Mbit/s half-duplex on thick coaxial cable;
- 10 Base 2 for 10 Mbit/s half-duplex on thin coaxial cable.



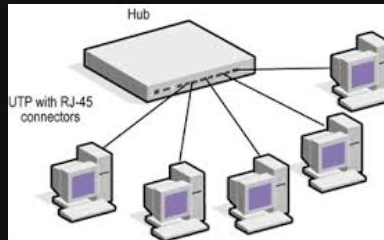
CSMA/CD

Carrier Sense Multiple Access with Collision Detection



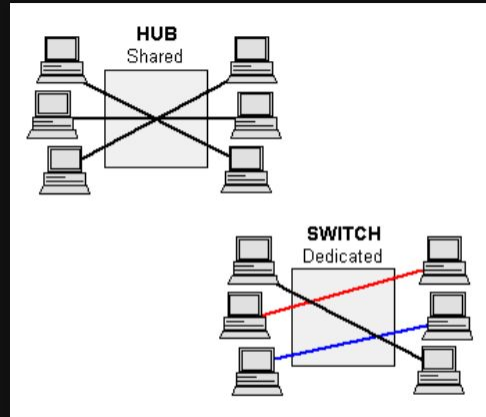
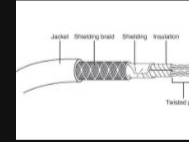
Ethernet Hub

- Replaces the coaxial bus by acting as a single network segment;
- Connects multiple Ethernet devices together;
- 10 Base-T for 10 Mbit/s half-duplex on unshielded twisted pair cabling.



Ethernet Switch

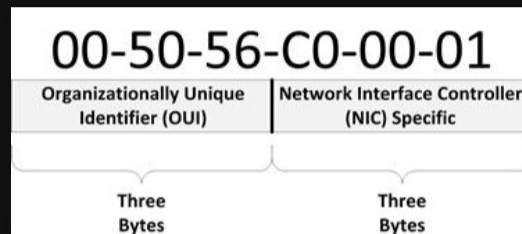
- Replaces shared medium with dedicated full duplex links;
- Prevents collisions;
- 100 Base-TX for 100 Mbit/s full-duplex on shielded twisted pair cabling.



Ethernet Address

A Media Access Control (MAC) address is used as a global unique identifier assigned to ethernet network interfaces for communications on the physical network segment

- 48-bit addressing scheme
- MAC address encodes the manufacturer identification number



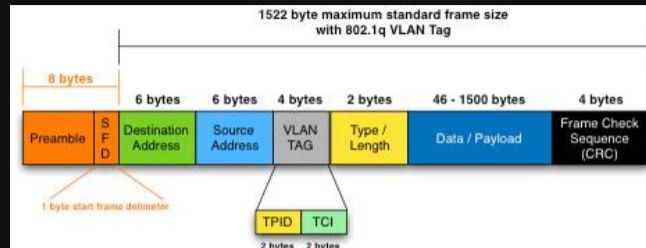
{ Exercise }

Find out your desktop MAC address and look for its manufacturer online

<http://macvendors.com>

Ethernet Frame

- Contains source and destination addresses;
- Up to 1500 bytes payload;
- CRC to check for errors.

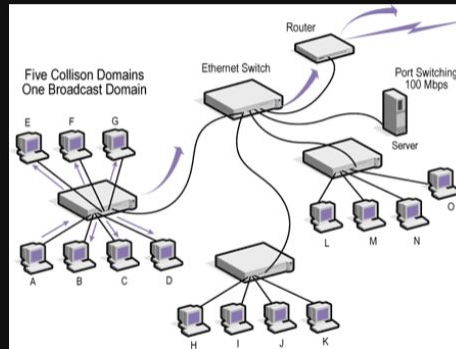


The Internet

Internet Architecture

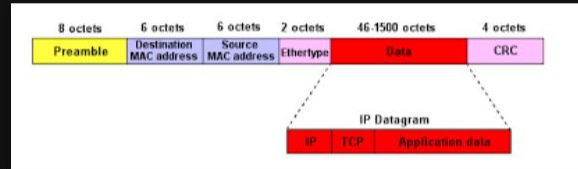
We have seen how machines connect to individual networks...
But how are networks interconnected to form an internetwork?

- We need computers that connect to multiple networks;
- We need these computers to forward packets between networks;
- Such computers are called Routers;
- Forwarding network traffic among networks is called Routing.



Internet Protocol

- Abstracts underlying LAN technology, making the internet appear to be a single uniform entity;
- Has the task of delivering packets from a source host to a destination host;
- IPv4 is the dominant protocol of the Internet;
- IPv6 is its successor.

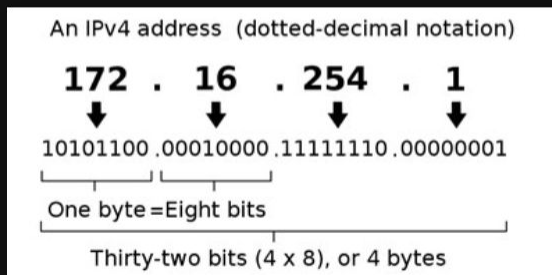


Internet Addresses

Each host on the internet is assigned a unique 32-bit internet address that is used in all communication with that host

IP addresses encode both a network and a host on that network. They do not specify an individual computer, but a connection to a network

If a host computer moves from one network to another, its IP address must change



{ Exercise }

Check IP addresses on the codecadets server using the ifconfig command

Subnet Masks

The parts of the IP address that are used as the network and host addresses are not fixed, so the network and host addresses cannot be determined unless you have more information.

This information is supplied in another 32-bit number called a subnet mask.

Class A	Network	Host	Host	Host
Subnet Mask	255	0	0	0

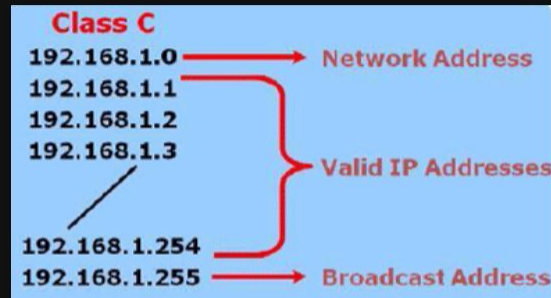
Class B	Network	Network	Host	Host
Subnet Mask	255	255	0	0

Class C	Network	Network	Network	Host
Subnet Mask	255	255	255	0

Network and Broadcast

The first address of a subnetwork is reserved for addressing the network itself

The last address of a subnetwork is reserved for broadcasting to all hosts on the network



{ Hands On }

Using an IP calculator such as ipcalc

Loopback Address

All addresses starting with 127 are reserved for testing and inter-process communication on the local machine. They are called loopback addresses and should never appear on the network

When a software program uses the loopback address as a destination, the operating system returns the data back without sending traffic across any network

There's no place like
127.0.0.1

Private Addresses

Private IPv4 address ranges are reserved for private networks and can not go out to the public internet

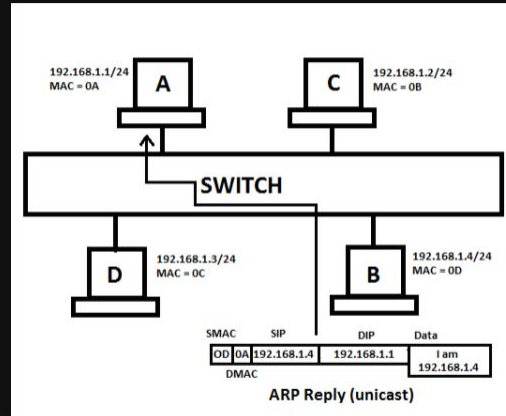
Private addresses can be translated to one or more public addresses by using a technique called NAT and/or PAT

Class	Private Address Ranges
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255
Loopback	127.0.0.0 – 127.255.255.255 (127.0.0.1)

Address Resolution Protocol

ARP is a low-level protocol that hides the underlying network physical addressing, which allows an arbitrary IP address assignment to each host

- Allows a host to find the physical address of a target host on the same physical network, given only the target IP address
- A broadcast ARP request is used to find the hardware address of target hosts
- To make ARP efficient, each host caches IP-to-physical address bindings



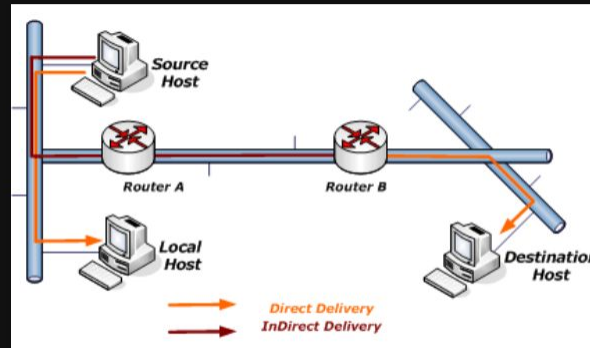
{ Exercise }

Print the arp table on the codecadets server using the arp command

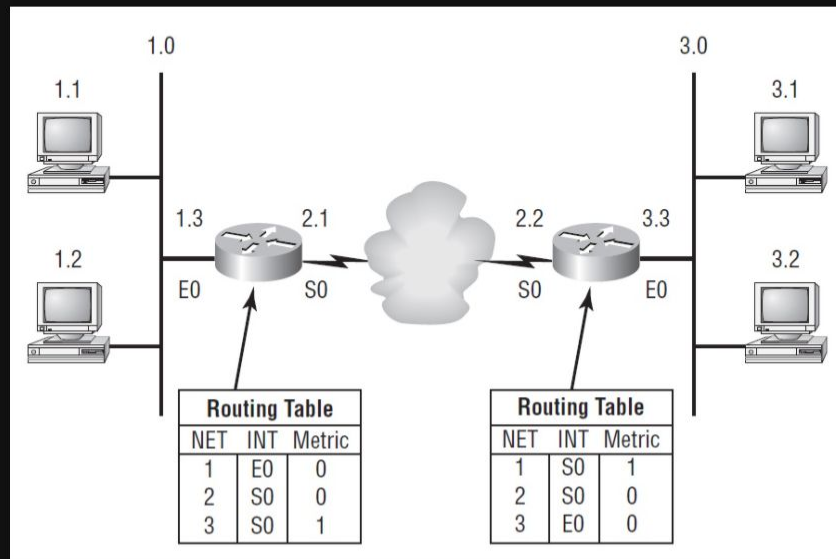
IP Routing

Routing is the mechanism that determines the path that data follows in order to travel across multiple networks from its source to its destination. Data is routed from its source to its destination through a series of routers, and across multiple networks

- frames destined to the same physical network use direct delivery
- frames destined to another physical network use indirect delivery
- each host has a default router or gateway responsible for forwarding traffic to other networks



Routing Tables



{ Hands On }

**Inspect and Manipulate the routing table on your server
using the route command:**

1. print the default route
2. add a new route for host 4.2.2.1 - requires root privileges
3. add a new route for network 4.2.0.0/16 - requires root privileges

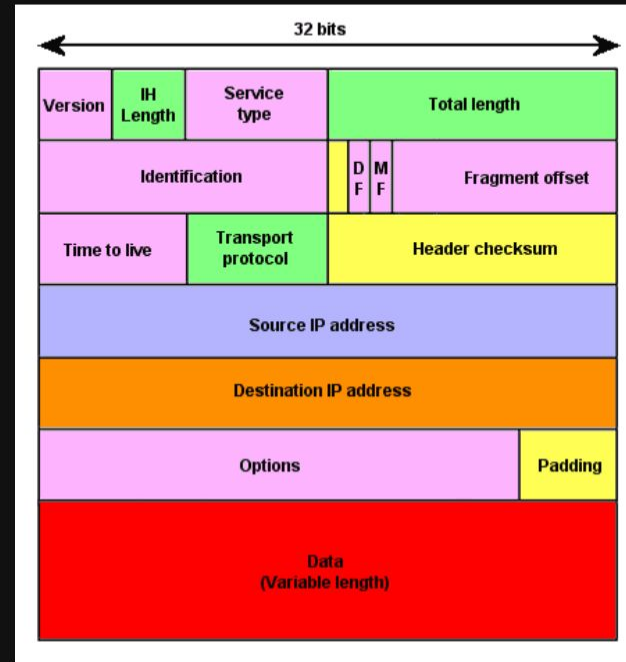
Connectionless Datagram Delivery

Unreliable, best-effort,
connectionless packet delivery
system

IP datagrams may be lost,
duplicated, delayed or delivered
out of order

Datagrams bigger than MTU will
be fragmented

IP fragments need to be
reassembled at the destination



{ Hands On }

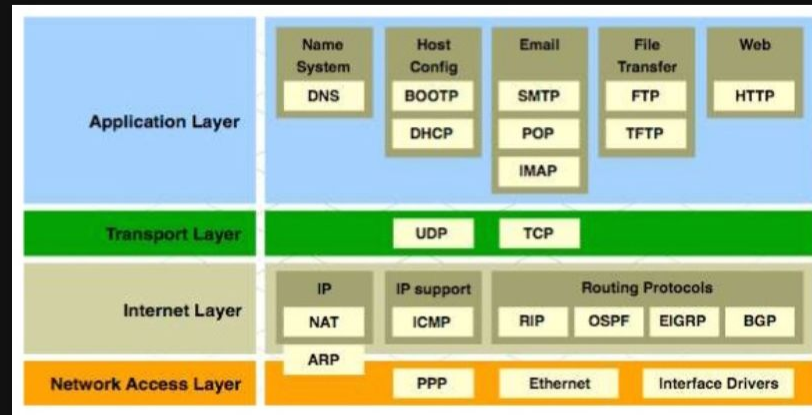
Alter the interface MTU on the codecadets server using the command ifconfig - requires root privileges

{ Hands On }

Display the IP statistics using the netstat command

TCP/IP

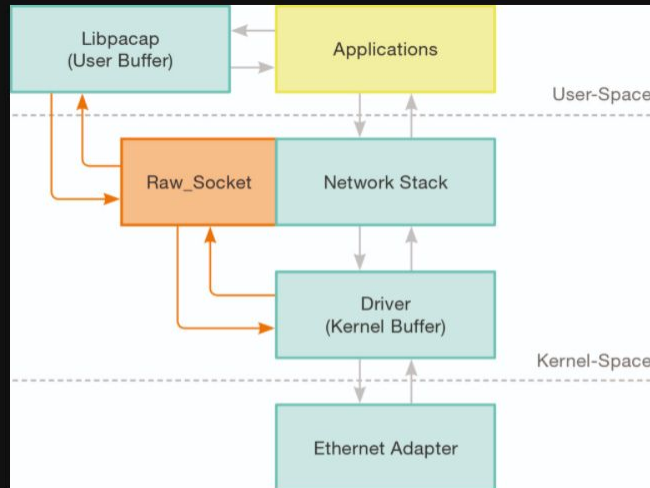
Acronym that is commonly used for the set of network protocols that compose the Internet Protocol suite



libpcap

Portable and system independent API for capturing network traffic

1. Written in C, but wrappers exist for many languages
2. Supports saving captured packets to a file
3. Supports reading files containing captured packets
4. Supports a filtering mechanism based on the architecture of the BSD packet filter
5. Applications include network statistics collection, security monitoring, network debugging, etc.



tcpdump

powerful command-line packet analyzer based on libpcap



- places the NIC in promiscuous mode
- may require root privileges, use sudo

```
// capture all packets on interface eth0 to screen
$ tcpdump -i eth0

// display captured packets in HEX and ASCII
$ tcpdump -XX -i eth0

// capture 10 packets up to 1500 bytes each on interface eth0, save them to file, filtering by
$ tcpdump -c 10 -i eth0 -s 1500 -w data.pcap host www.academiadecodigo.org

// read packets from a previously saved file, be very verbose about it
$ tcpdump -vvv -r data.pcap
```

man tcpdump for additional options/features

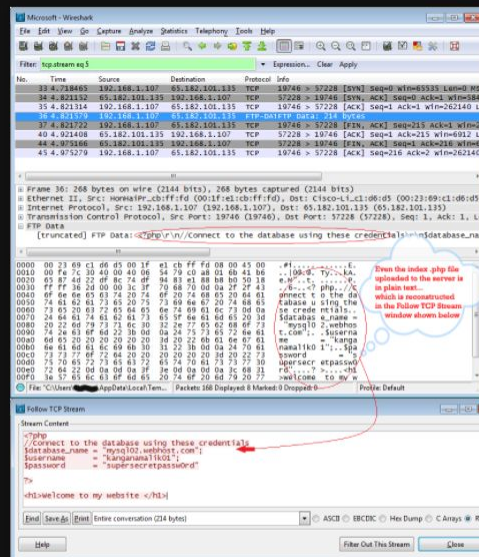
{ Exercise }

Generate, capture and display network traffic on the
codecadets server using tcpdump

Wireshark



Very similar to tcpdump, but has a graphical front-end
and integrated sorting and filtering options



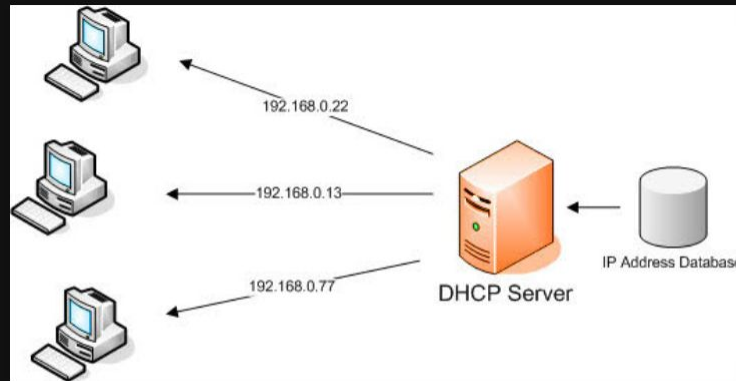
{ Exercise }

Generate, capture and analyse network traffic on your desktop

traffic capture requires root privileges, capture on the server and transfer with scp to the desktop

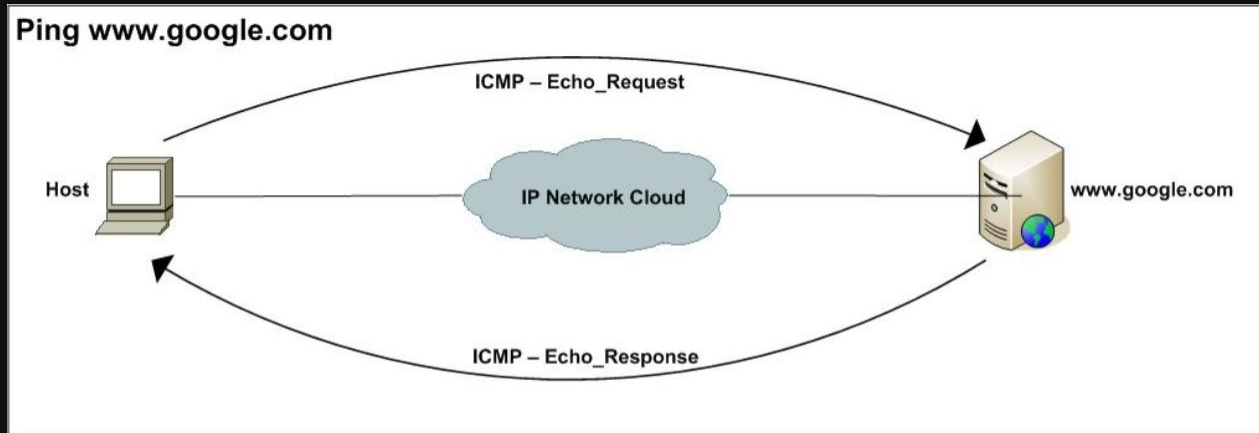
Dynamic Host Configuration Protocol

With DHCP, computers request IP addresses and networking parameters automatically from a DHCP server, reducing the need for a network administrator or a user to configure these settings manually



Internet Control Message Protocol

ICMP is one of the main protocols of the Internet Protocol Suite. It is used by network devices to send error messages indicating, for example, that a requested service is not available or that a host or router could not be reached



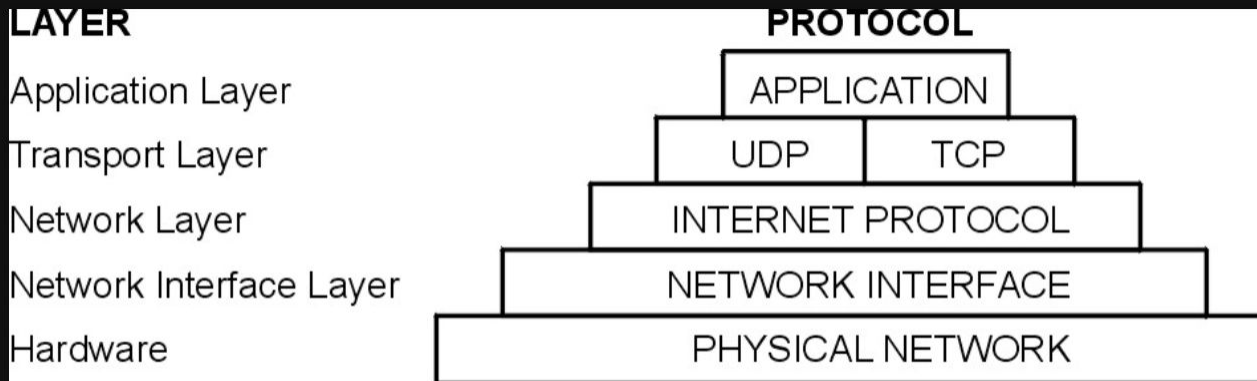
{ Exercise }

Perform reachability tests to well known websites on the codecadets server using the ping and tracert commands

Transport Layer

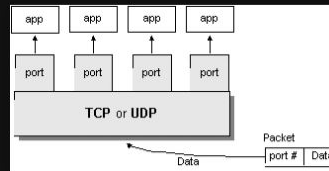
Provides end-to-end communication from one application program to another

- may regulate flow of information
- may provide reliable transport
- must accept data from multiple user programs
- programs within a host are addressed by port number

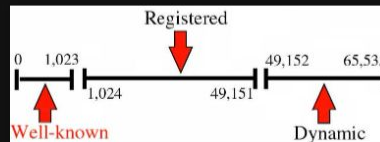


Ports

A port number is a way to identify a specific process to which an Internet or other network message is to be forwarded when it arrives at a server



Lower port numbers are reserved for standardized protocols, higher port numbers are free for use



Port Numbers

Label on Column	Service Name	UDP and TCP Port Numbers Included
DNS	Domain Name Service – UDP	UDP 53
DNS TCP	Domain Name Service – TCP	TCP 53
HTTP	Web	TCP 80
HTTPS	Secure Web (SSL)	TCP 443
SMTP	Simple Mail Transport	TCP 25
POP	Post Office Protocol	TCP 109, 110
SNMP	Simple Network Management	TCP 161,162 UDP 161,162
TELNET	Telnet Terminal	TCP 23
FTP	File Transfer Protocol	TCP 20,21
SSH	Secure Shell (terminal)	TCP 22
AFP IP	Apple File Protocol/IP	TCP 447, 548

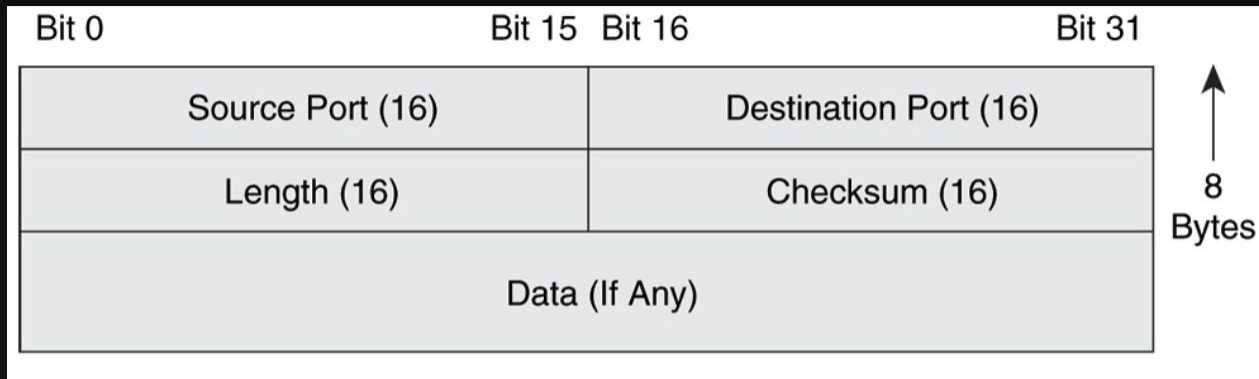
```
// dump all well-known ports to screen  
$ cat /etc/services
```

```
// dump all http like protocol ports to screen  
$ cat /etc/services | grep http
```

User Datagram Protocol

UDP provides an unreliable connectionless delivery service using IP to transport messages between machines

It uses IP to carry messages, but adds the ability to distinguish among multiple destinations within a given host computer

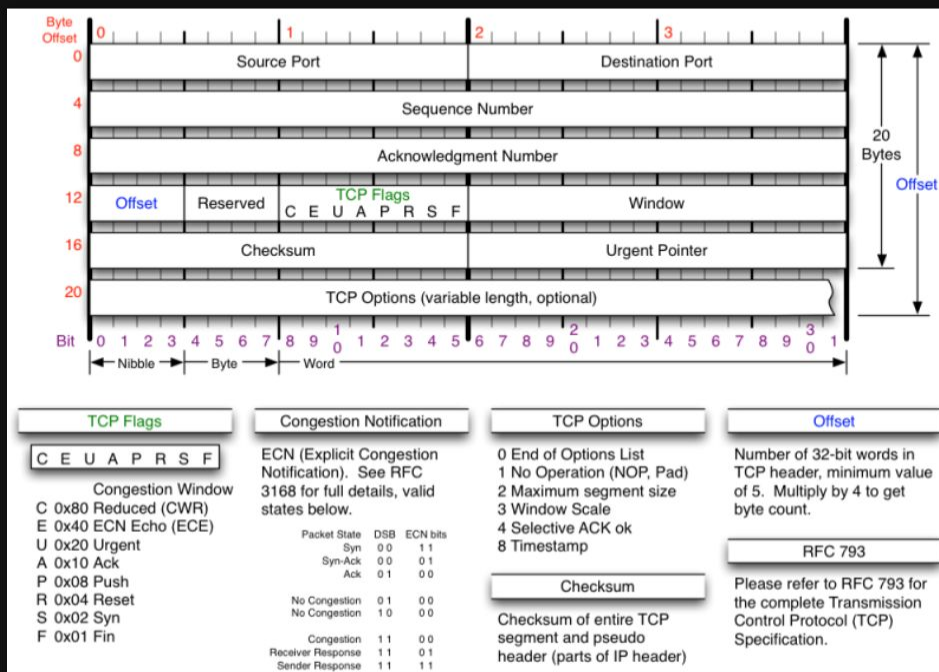


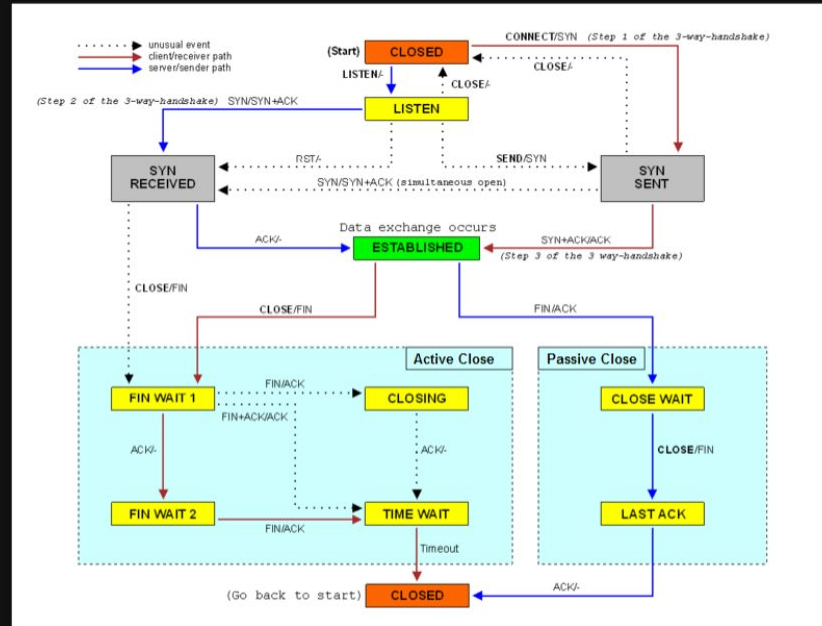
Transmission Control Protocol

TCP provides a reliable full duplex connection between two machines, allowing them to exchange large volumes of data efficiently

- Applications request a connection to the destination, which is used to transfer data;
- Applications can handle multiple tcp connections simultaneously;
- Guarantees that data will be delivered without loss, duplication or transmission errors;
- Endpoints can exchange data in both directions simultaneously;
- Guarantees reliable, synchronized startup between endpoints;
- Guarantees delivery of all data after endpoint shutdown;
- Provides flow control, allowing systems of varying speeds to communicate.

TCP Header





{ Exercise }

Check TCP connection states on the codecadets server
with the netstat command

Socket API

Provided by the operating system, allowing application software to use network services without being concerned with low-level communication details

Sockets are characterized by:

- Transport protocol, which can be UDP, TCP or raw data
- Local IP address
- Local Port number
- Remote IP address
- Remote Port number

{ Hands On }

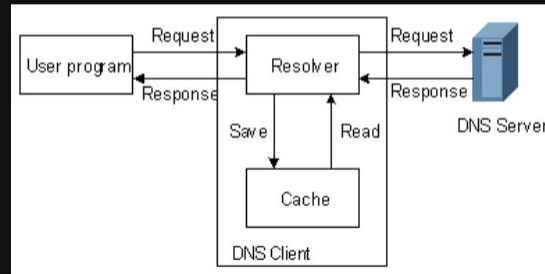
Check for listening and established sockets on the codecadets server with the netstat command

Domain Name System

The DNS is a hierarchical distributed naming system for computers, services, or any resource connected to the Internet or a private network. It associates various information with domain names assigned to each of the participating entities

We can query DNS for:

- address for domain name
- address of dns servers for domain
- address of mail servers for domain

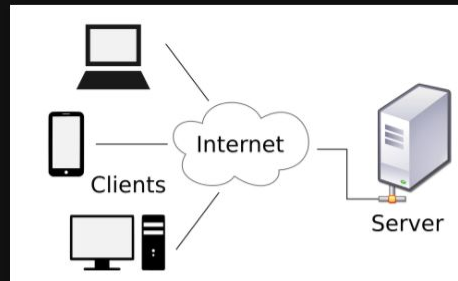


{ Hands On }

resolve hostnames, nameservers and mail servers on the codecadets server using the host command

Client-server model

Architecture of a computer network in which many clients request a service from a centralized server (host computer). Servers wait for requests to arrive from clients and then respond to them.



Usually servers are more powerful machines located elsewhere on the network, but it is possible for the client and the server to reside on the same host

Netcat

The TCP/IP Swiss Army knife



- Listen on arbitrary TCP and UDP ports
- Establish TCP connections
- Send UDP packets
- Perform port scanning (you might want to check nmap for some serious security scanning)

{ Exercise }

Manually establish tcp/udp sockets with netcat
Capture and decode traffic

Network Programming in Java

java.net.InetAddress

High-level representation of an IP address and used by most of the other networking classes

Capable of performing DNS resolutions, storing results in a local cache and testing for host reachability

{ Exercise }

Host name resolution and reachability

Sockets

Different processes can communicate with each other across networks by means of sockets. Java implements both TCP and UDP sockets, allowing the programmer to treat a network connection as just another stream.

`java.net.DatagramSocket` implements UDP sockets, capable of sending and receiving UDP datagrams to/from a specified host and port

`java.net.Socket` implements TCP client sockets, capable of connecting to a specified server and port

`java.net.ServerSocket` implements TCP server sockets, used to wait for requests to come in over the network.

UDP Socket

Steps for sending/receiving to/from udp socket:

```
String hostName = args[0];
int portNumber = Integer.parseInt(args[1]);

// STEP2: Create send and receive buffers
byte[] sendBuffer = new byte[1024];
byte[] recvBuffer = new byte[1024];

// STEP3: Open a UDP (datagram) socket
DatagramSocket socket = new DatagramSocket(portNumber);

// STEP4.1: Create and receive UDP datagram packet from the socket
DatagramPacket receivePacket = new DatagramPacket(recvBuffer, recvBuffer.length);
socket.receive(receivePacket); // blocks while packet not received

// STEP4.2: Create and send UDP datagram packet from the socket
DatagramPacket sendPacket = new DatagramPacket(sendBuffer,
    sendBuffer.length, InetAddress.getByName(hostName), portNumber);
socket.send(sendPacket);

// STEP5: Close the socket
socket.close();
```

{ Exercise }

Implement UDP UPPERCASE server and client classes

TCP Client Socket

Steps for reading/writing to/from client socket:

```
// STEP1: Get the host and the port from the command-line
String hostName = args[0];
int portNumber = Integer.parseInt(args[1]);

// STEP2: Open a client socket, blocking while connecting to the server
Socket clientSocket = new Socket(hostName, portNumber);

// STEP3: Setup input and output streams
PrintWriter out = new PrintWriter(clientSocket.getOutputStream(), true);
BufferedReader in = new BufferedReader(new InputStreamReader(clientSocket.getInputStream()));

// STEP4: Read from/write to the stream
// STEP5: Close the streams
// STEP6: Close the sockets
```

TCP Server Socket

Steps for reading/writing to/from server socket:

```
// STEP1: Get parameters from command line arguments
int portNumber = Integer.parseInt(args[0]);

// STEP2: Bind to local port and block while waiting for client connections
ServerSocket serverSocket = new ServerSocket(portNumber);
Socket clientSocket = serverSocket.accept();

// STEP3: Setup input and output streams
PrintWriter out = new PrintWriter(clientSocket.getOutputStream(), true);
BufferedReader in = new BufferedReader(new InputStreamReader(clientSocket.getInputStream()));

// STEP4: Read from/write to the stream
// STEP5: Close the streams
// STEP6: Close the sockets
```

{ Exercise }

Chat server and client (support for 1 client only)